

Ethics in Nanotechnology

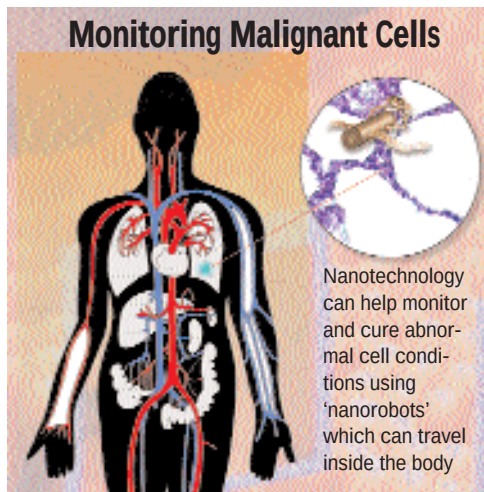
Nanotechnology deals with the most basic assembly of matter. This is like delivering the power of creation into human hands.

Consider the concept of self-replicating robots that traverse the human body providing all sorts of cures. The only self-replicating mechanisms we know of are the reproductive mechanisms of living cells. Even in their case, we are not really clear about the intricacies involved. So it is quite possible that accidental damage could be caused by unleashing something which we can't control, as our understanding of the complexities involved is limited.

Chances of a deliberate misuse of this technology are even higher. One only has to think of the immense destructive potential that biotechnology has brought about through weapons of biological warfare.

Some experts have called for banning nanotechnology, but this will be like killing one of the greatest developments in technology even before it's happened. This is where ethics takes the centre stage.

The Foresight Institute has laid down a set of guidelines for nanotechnology research and development. The guidelines are reviewed yearly and new guidelines are formulated as research brings up newer issues.



ory capacity in a matchbox-sized drive. Big Blue's next step is to make gigabyte drives as tiny as one square centimetre. This means that music aficionados can store their entire music collection on something the size of a watch a few years from now. There are also plans to build a smart version of Millipede. This will have electronics coupled to the Microdrive so that the whole package can act as an independent system on a chip.

That was just the tip of the iceberg. Molecular switches and transistors were first developed in 1998. These breakthroughs are showing a path on which a completely new kind of computer architecture is being developed. This approach makes use of objects called Quantum dots for their operation. (See Box: *Quantum dot cell operation for more information on this new architecture.*)

The main advantage of using approaches such as the Quantum dot cell, other than the small sizes, is the power consumption. In an era where we are strapped for power, nano-powered devices will be a big help as they consume next to nothing in terms of power.

Here, there, and everywhere

Materials technology is another major area which will be affected by Nanotechnology. A nanotube is one such innovation which can change almost all the areas that we are familiar with. These are single walled molecules made from buckyballs. We'll see what buckyballs are in detail later on. The main advantage of using these nanotubes is that it is possible to control the way these crystals are developed for particular applications. Electrical and

other properties of materials made using nanotubes can be made to fit precise specifications. This makes it possible to use them for practical applications, from clothing to superconductivity applications to spacecraft design.

Nanotechnology has been in the market for a while now. Toyota and Nike use Nanotechnology in some of their products. Toyota started using a nanocomposite material early in the 1990s in their engines. This material, made out of reinforced nylon, withstands high engine temperatures and is cheaper than other forms of similar plastic. The Nike trainers, which are so commonly available, make use of nanoengineered cushioning material in the soles of some Nike shoes.

Miracle drugs

We all dread operations, but we might not have to in the near future, not with Nanotechnology.

The concept here is Microencapsulation, a Nanotechnology concept, which will help doctors control precisely the rate at which medicines are supplied to the patient's body. One of the major medical breakthroughs in this area is a composite structure of carbon called buckyballs, or C60 molecules. Buckyballs were discovered by Richard Smalley of Rice University in Houston for which he got the Nobel Prize in 1985. The main advantage of using buckyballs for medical purposes is that they are extremely small and are non-toxic. These spherical particles are a nanometer long and very smooth. They are easily excreted by the body, which makes them perfect as drug delivery mechanisms. Researchers have developed a family of structures known as Fullerenes based on buckyballs. These are tube-like structures which could turn the entire medical field around because of their immense potential in the way treatment is administered.

Using Fullerenes, medicines could be delivered to the body orally and the body then eliminates the Fullerenes without any side effects. It is possible to attach the needed drugs on the buckyball structure as required for the particular disease. This is much easier and effective than the conventional capsule approach. In capsules, a mixture of drugs is delivered into the body, a major part of which is eliminated by the body. When using mechanisms like buckyballs, it is easy to ensure that they are tailor-made to deal with the specific cell disorder that the disease causes.

Drugs that make use of buckyballs for the

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